

The Vicsek-graph Conjecture C of arXiv:2502.10837 is answered by arXiv:2606.05475

Literature-status note

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Status. The later Theorem 1.3 proves source Conjecture C in a stronger form and disproves the companion extension proposed immediately after it. It gives a complete classification away from one critical exponent; equality remains open. This is a literature implication, not a new proof.

1 The source question

For a Vicsek graph with diffusion escape exponent β , arXiv:2502.10837v3 compares

$$(R_{p,\gamma}) : \quad \|\nabla f\|_p \leq C \|\Delta^\gamma f\|_p, \quad (RR_{p,\gamma}) : \quad \|\Delta^\gamma f\|_p \leq C \|\nabla f\|_p.$$

Conjecture C on PDF page 8 states that, for $p > 2$, $(RR_{p,\gamma})$ holds whenever

$$\gamma > \theta(p) := \frac{1}{\beta} + \frac{1}{p} \left(2 - \frac{2}{\beta} \right). \quad (1)$$

The following comment proposes the companion assertion that, for $1 < p < 2$, $(R_{p,\gamma})$ holds whenever $\gamma < \theta(p)$, and calls equality open [1].

2 The later classification and the exact implication

For a D -dimensional Vicsek graph one has $\beta = D + 1$. Theorem 1.3 of arXiv:2606.05475v2 (PDF page 5) sets

$$\gamma_*(p) = \frac{1}{D+1} + \frac{1}{p} \left(1 - \frac{2}{D+1} \right) = \frac{1}{\beta} + \frac{1}{p} \left(1 - \frac{2}{\beta} \right) \quad (2)$$

and proves, for every $1 < p < \infty$ and away from equality,

	$(R_{p,\gamma})$	$(RR_{p,\gamma})$
$0 < \gamma < \gamma_*(p)$	holds	fails
$\gamma_*(p) < \gamma < 1$	fails	holds.

The paper explicitly says that its main goal is to prove the Devyver–Russ reverse-transform conjecture in the discrete Vicsek setting [2].

The source and later thresholds satisfy the elementary identity

$$\theta(p) - \gamma_*(p) = \frac{1}{p} > 0.$$

Consequently $\gamma > \theta(p)$ implies $\gamma > \gamma_*(p)$, so the later theorem proves source Conjecture C, in fact on the strictly larger range $\gamma > \gamma_*(p)$.

The proposed companion extension is false. Since $\gamma_*(p) < 1$, for every $1 < p < 2$ one can choose

$$\gamma_*(p) < \gamma < \min\{\theta(p), 1\}.$$

This γ satisfies the proposed hypothesis $\gamma < \theta(p)$, but Theorem 1.3 says that $(R_{p,\gamma})$ fails.

3 Remaining scope

The precise remaining question is the single line $\gamma = \gamma_*(p)$, not the larger threshold $\theta(p)$ printed in the 2025 proposal. The 2026 paper expressly leaves that critical case open. A separate lane packet for arXiv:2606.05475 records new partial endpoint information; it does not claim the endpoint classification.

Thus the defensible status is

source Conjecture C: yes; proposed companion: no; critical line $\gamma = \gamma_*(p)$: open.

References

- [1] J. Feneuil, *In spaces with a slow diffusion, the Riesz transform is unbounded on L^p , $p \in (2, \infty)$* , arXiv:2502.10837v3 (2025), Conjecture C.
- [2] J. Feneuil, *Reverse inequalities for super-Riesz transforms on graphs with a slow diffusion*, arXiv:2606.05475v2 (2026), Theorem 1.3.